ISRO SMART INDIA HACKATHON 2025 · DTIE PROJECT

LunaBot

Explore. Perceive. Autonomously.

Autonomous Navigation for Lunar Habitats

TEAM

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The Challenge

Communication Delay
A 1.3 s round trip makes real-time remote control impractical.

Extreme Lighting
Permanent shadows and harsh glare in the same mission area.

Loose Regolith
Powdery soil causes slippage and unpredictable traction.

Who We Are Solving This For

Astronauts on EVA. Need a rover that does not demand attention during surface walks, where oxygen and power are limited.

Ground Control Operators. A 1.3 s delay rules out joysticks; they issue high-level intent like “patrol habitat”.

Habitat Safety Teams. Need continuous perimeter awareness without crew time spent on manual patrols.

Why Standard Solutions Fall Short

Vision-only fails in shadows. Cameras break down in permanently shadowed craters.

Slope checks miss regolith. Geometry ignores how wheels sink and slip in loose soil.

Human-in-the-loop. Real-time correction breaks under communication delay.

Orbital maps are too coarse. They cannot resolve wheel-level rocks, ruts or soft patches.

Today’s rovers force a choice: slow, safe, supervised movement or fast, risky, unsupervised movement. Neither is acceptable.

Our Solution: The LunaBot Stack

Multi-Modal Perception
Stereo camera + thermal + depth sensing works in lit and shadowed terrain.

Smart Terrain Scoring
Slope, surface roughness and lighting fused into one traversability score.

True Autonomy
A behavior-tree state machine runs with no operator in the loop.

How It Works: Closed-Loop Intelligence

1 **Waypoints**
Ground control sends high-level goals, e.g. “patrol habitat”.

→ 2 **Perception**
Orbital maps fused with live sensor data build the terrain model.

→ 3 **Cost Map**
A composite cost map steers away from high-slip and shadowed hazards.

→ 4 **Decision**
Onboard avoidance and recovery; every decision is logged for audit.

Idea Generation: Options Considered

Pure vision / camera SLAM · GNSS + orbital-map following only · Teleoperation with delay buffering · Swarm / multi-rover relay

Chosen: multi-sensor fusion with onboard cost mapping
Camera + thermal + LiDAR/ultrasound with terrain scoring and autonomous decisions. This became the LunaBot concept.

Real-World Tasks

Autonomous Docking
Self-directed docking at charging stations.

Habitat Patrols
Scheduled perimeter patrols on a timetable.

EVA Support
“Follow-me” mobile assistant on surface walks.

Emergency Return
Fastest-safe-return routing under power pressure.

Rough Prototype

FULL CONCEPT: HARDWARE

Chassis · LiDAR (ultrasound used) · Stereo camera · Thermal sensor · Ultrasound array

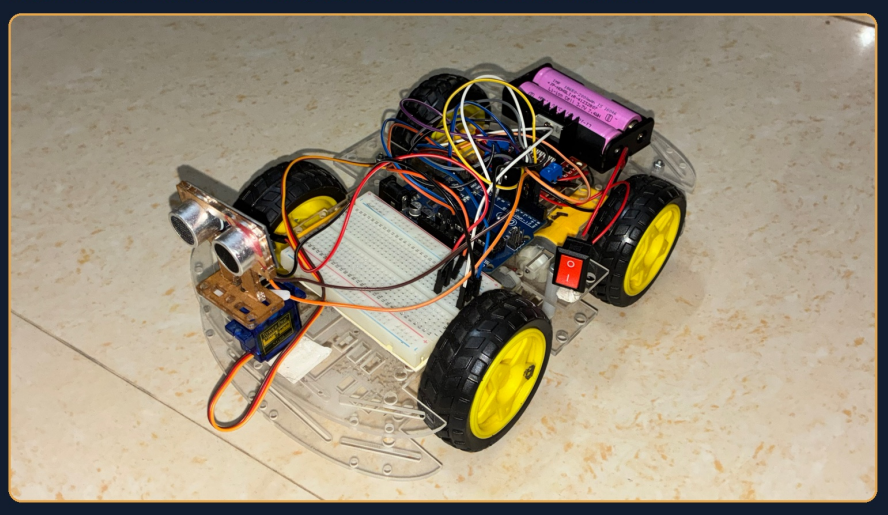
SOFTWARE STACK

SLAM localization · Sensor fusion pipeline · Cost-map generator · Behavior-tree decision layer

TEST ENVIRONMENT

Controlled testbed: simulated regolith arena / dark-lit test zone, for validation before final integration.

WORKING PROTOTYPE: OBSTACLE-AVOIDING ROVER



4WD chassis, 4 DC motors · Arduino Uno · HC-SR04 ultrasonic (stands in for LiDAR) · L298N motor driver · 2x18650 pack (7.4 V)

Read distance → > 25 cm: drive forward → else: stop, reverse, turn